BAIM 4120 · CLASS 02 · WHITE BELT

FROM ALTERYX TO PYTHON

Rebuilding today’s Alteryx workflow in a Jupyter notebook with pandas

Leeds School of Business · University of Colorado Boulder · github.com/iportilla/white-belt

TODAY

BY THE END OF CLASS YOU CAN…

1. Load a CSV into pandas and **browse** it like the Alteryx Browse tool
2. Explain why **data types** matter, and convert columns to the right type
3. **Filter** rows with a boolean mask into True and False outputs
4. **Sort**, rename, drop and **save** data
5. Describe any workflow as **Extract** → **Transform** → **Load**

THE DATA

PHONE APP EVENTS FROM KAGGLE TALKINGDATA

Column	What it means
event_id	Unique ID of each event
device_id	ID of the phone (can be negative)
timestamp	When the app was used
longitude	Where: east–west
latitude	Where: north–south

~195 MB

Millions of rows: too big for Excel to be comfortable

Watch for: a location of **0, 0**. It isn't a real place.

THE BIG PICTURE

EVERY WORKFLOW IS ETL

EXTRACT

Read the CSV in

`pd.read_csv()`



TRANSFORM

Fix types, filter, sort

`astype() · df[mask]`



LOAD

Save the clean result

`to_csv()`

Alteryx and Python are two ways of writing the same three steps.

SAME WORKFLOW, NEW TOOL

EVERY ALTERYX TOOL HAS A PANDAS TWIN

Alteryx tool	pandas	Lab part
Input Data	pd.read_csv()	A
Browse	head() · info() · dtypes	A
Select: types	astype() · pd.to_datetime()	B
Select: drop / rename	drop() · rename()	C
Filter + Sort	df[mask] · sort_values()	D
Output Data	to_csv(index=False)	E

THE BIG IDEA

WHICH IS BIGGER: 999 OR 1000?

COMPARED AS TEXT

“999”

Wrong. It looks at the first character, and 9 beats 1.

COMPARED AS NUMBERS

1000

Right. Numbers compare by value.

SET TYPES BEFORE YOU SORT, COMPARE OR CALCULATE.

FILTERING

A MASK IS THE FILTER TOOL'S T AND F

ALL ROWS

```
mask = df["lon"] == 0
```



T `df[mask]`
lon is 0: location unknown

F `df[~mask]`
lon isn't 0: real locations (~ means NOT)

Combining conditions? Use **&** and **|**, with each one in parentheses: **(a == 0) & (b != 0)**

THE NOTEBOOK

TWO SECTIONS: TRY IT, THEN CHECK IT

1 · STARTER

Explanations, then code cells marked # **TODO**. Fill them in and answer the 憊 questions.

2 · SOLUTION

Finished code, with the reasons it works. Open it **after** you try each part.

A · Browse	B · Types	C · Drop	D · Filter	E · Save

THE LAB

HOW TODAY'S LAB WORKS

60
minutes, in class

Pairs
driver and navigator, switch at Part C

20
points

Get it: github.com/iportilla/white-belt (clone it, or Code → Download ZIP)

Do it: every # TODO and 憊 cell in the Starter

Hand it in: only your .ipynb on Canvas, by the end of class

TAKE AWAY

SET TYPES FIRST.
CHECK THAT T + F = TOTAL.
SAVE WITH INDEX=FALSE.

Stuck? Read the README, then ask your neighbour, then post in **#ba-homework-help** on Slack.